SUSTAINABLE DESIGN ENGINEERING

submission 2



Problems & Industry Direction

Industrial Changes

How might we enable modular upgrades in airline-specific long-haul economy seats to prolong their lifespan, allowing components to be quickly refreshed and serviced, and keeping end-of-life recycling straightforward?

Visual Product

How might we redesign long-haul economy seats with modular branding interfaces, for enhanced user perceived value, which can be quickly cleaned or refreshed and operate within a fully circular system?

Airline Industry Insights

Pain points & key insights identified within each sector.

Product · Industry · System · User

- **Heavy branding reduces serviceability and limits customisation**, leading to negative economic and environmental impacts.
- **Brand differentiation remains essential** for passenger experience and stakeholder value.
- The primary market driver is **reducing weight** to control costs and lower CO_2 emissions, as seats add weight, thereby increasing fuel burn, adding up over time.
- **A fastener-less approach** could cut seat weight and deliver lasting fuel efficiency gains. Current seats have a complicated architecture.
- Seats have **not been designed for disassembly,** and as such waste the precious virgin aluminium when stored or disposed of. No instructions provided either.
- There is **little collaboration between airlines and seat manufacturers**, leaving much to be desired in terms of design.
- **Seat covers drive high costs** and hinder circularity due to need for frequent cleaning and replacement, highlighting the **need for easy cleaning processes** and recyclability.
- Sustainability should enable **gradual market adoption** to avoid the environmental impact of sudden large-scale overhauls.
- **Regulations** by aviation bodies, particularly the FAA, have **bottlenecked production** and supply of seats to airlines, whilst **limiting changes to their design.**
- Airlines are slow to adopt new technologies within interiors, until a competitor introduces a change, in which case they flock to keep up with the market.
- As delays and costs go up, airlines are **cramming more seats** in a bid to increase turnover and profits, however this comes at a **cost of comfort** to passengers.
- Privacy is an issue in economy, as you are with a stranger for up to 17 hours, meaning customers may feel tense or not as comfortable as they could be otherwise.

Future of the airline industry- context

In line with SDG's target of achieving **net zero by 2050**, many airlines have pledged efforts to meet this target, primarily through SAF which airlines like Virgin Atlantic and Emirates have been key testers for, as well as **route optimisation**. Beyond this however, there has been **little incentive** to change other areas, such as seats, recycling or regulations, other than for **responses to public demand** for visible improvements.

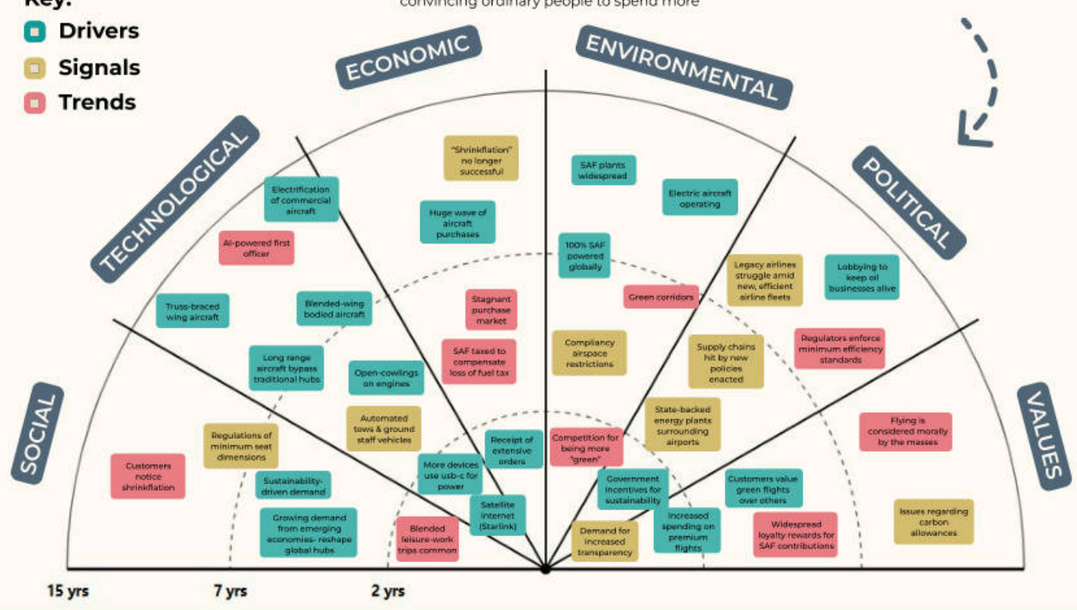
Whilst the industry has been advancing in development of electric-powered aircraft, composite bodies like in the 787, morphing wings in the A350, and folding wingtips in the 777X, we are still an **estimated two decades away** from full-scale deployment of long-haul electric commercial aircraft.

Thus, **sub-assemblies** and the systems within which aircraft operate in should be given attention, as there is vast room for improvement in often overlooked areas of aircraft operation.

Future of the Aviation Industry- exploration

STEEPV Focus wheel on the industry's future

The airline industry is spiked, and reactive to social trends and government policies, and has rapidly expanded in recent years. Traditional business models are being scrapped in favour of convincing ordinary people to spend more





Circadian Lighting



Stacked Seating[1]



Blended Wing [2]

Examples of innovations now, soon, and in the future within the industry

Focus:

Hard Product (seat) - Redesign components of a seat for brand modularity, considering sustainability & comfort, and to facilitate implementation of improved industry-wide systems.

Soft Product (systems)- Proactive rather than reactive- predict market direction and respond accordingly. Allow sustainable processes to be implemented more easily than currently. Valorisation implementations and shift the industry's current approach to disposal of parts. Improve the general flight experience in economy.

Next steps: identify key users and demographics within the industry.

Key Users

Meet some frequent flyers:

Matt: Operations Manager



Age: 47

Matt travels from London to Singapore 3 times a year, as well as making yearly trips to the USA.

He **prioritises comfort** over anything else, as he needs to work and be refreshed when he arrives. As flights are paid by his office, **sustainability is not a factor** in his booking decisions.

"It's hard enough not flying business class...and still need to get right to work when I land"



Ashjan: Creative Consultant



Age: 27

Ashjan travels regularly within south-east Asia and Oceania, as well as monthly trips to Europe and the Middle East for consultancy work.

As a freelancer, she pays for flights herself, electing to f**ly economy to reduce costs**. But the frequent flights dry her skin and **space is tight** for work.

"If I could just have a second screen on my flights I could get so much more work done"



Hamza: Holiday-goer



Age: 20

Hamza is a student in the UK, and goes on holiday abroad once or twice a year, with family.

Most of his outbound flights are due east, and his flights average to over 6 hours each. As such, he arrives the next morning, **tired and groggy** at the start of his holiday.

"I love going on holiday, but I feel like I lose 2-3 days just because of travel time and tiredness. And I hate when I book a window seat and the window isn't aligned"



Alexia: Flight Attendant



Age: 31

Alexia moved to Dubai to work with Emirates as part of the cabin crew. As such, she flies daily for her job, and makes use of corporate benefits to get discounts on her own trips.

Unless on an ultra-long flight, she **cannot use the cabin beds**, and must use the **small jump seats**.

"I love my job, I love flying. But it is demanding and long. And to sit in the jump seat after standing for so long. I wish more planes had crew rest areas, and space to stretch"



Systems Pain Points

Cabin Crew

Cabin crew have to stand for hours, and on the shorter end of long-haul flights, they only get a bit of rest in an uncomfortable jump seat. If this seat was made like a regular economy seat, it could ease operations and allow for better service by the cabin crew, which contributes to Skytrax ratings and thus brand imaging.


[3]

Maintenance Staff



[5]

Following on from the teardown of our seats, the labour costs for disassembly outweigh the profits from scrapping the materials. This has led to warehouses full of old seats, preventing recycling and airlines losing more money.

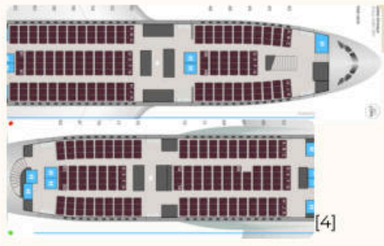
A laptop company called framework uses QR codes on parts inside their computers, linking to video guides for replacing them. This could be adopted within seat frames to speed up the disassembly process, encouraging airlines to recycle their seats whilst saving them money, or to downcycle the T3 and T6 aluminium to automotive parts.

Work-Space

Restricted space due to the crammed nature of economy class. Tray tables already extend far out, but screen real estate is not yet well utilised. Potential for screen extension from device for both work and entertainment- stream from your own device. British Airways is trialling "Bring your own device".



Seat Selection


[4]

Windows are often misaligned to seats- they should be better mapped to the aircraft, or at least a better seat selection tool made available, with a similar view to the Aerolopa seat-maps, with a disclaimer to avoid legal issues when aircraft swaps occur.

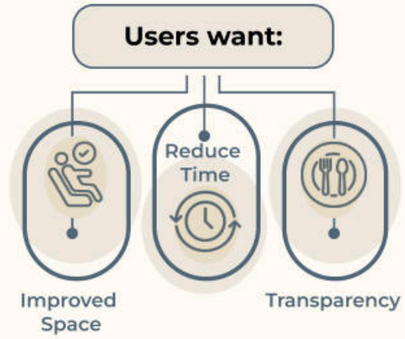
Airlines market their best product, when in reality, it is only typically in 20-40% of their aircraft. Seat widths vary per aircraft too.

Sustainable footprint awareness

Estimated CO_2 emissions are available, but not directly provided. Airlines could include them on user profiles to enhance awareness.

Industry-wide rewards for spending on SAF contributions. British Airways does this, awarding avios & tier points, but more should follow.

Users want:



Key takeaway from systems: The current state of the industry is not as well optimised as it could be from a sustainable perspective. There is much room for improvement at the systems level which can have an equally large impact as seat re-development.

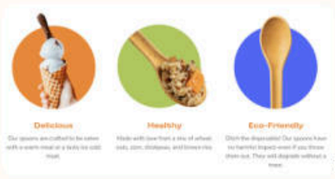
Next steps: Research key stakeholders within this sustainable initiative

Stakeholders & System Analysis

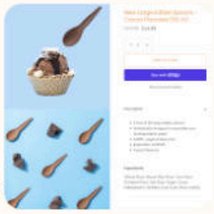
Stakeholder Initiatives

British airways is losing its prestige in economy, particularly with the quality of food presentation and packaging, opting for bamboo utensil, packets and no wrapped blankets. Wooden utensils still get thrown out though, and are hard to recycle with food and grease embedded in them.

A similar product to “IncredibleEats” could be implemented, whereby utensils are entirely edible, work in hot and cold perishables, and can be flavoured to match the food item. This is essentially no-waste, will not add to weight as it can be considered part of the meal, and could improve the food situation in the eyes of the consumer.



[6]



Civil Aviation Authority

FEDERAL AVIATION ADMINISTRATION

EASA European Aviation Safety Agency

Airline regulators: CAA, FAA, EASA: In order to speed up the certification process and introduce the concept of a second-hand certified market for airline seats, we need to liaise with governing bodies for approval and to set the work in motion.

BRITISH AIRWAYS

Airline: British Airways. As the flag carrier of the UK, they will be one of the stakeholders in this initiative to pioneer the industry, whilst setting an example in global sustainability to help meet the 2050 net-zero goal without compromise.
Use British Airways as the pilot airline in order to flesh out the concepts, but also provides a competitive edge, so it should favour both parties.

Gen Phoenix [7]

GenPhoenix - large-scale leather and fabrics recycling for over 15 years, and already proven collaboration with Southwest and AirBaltic. Sourcing the cosmetic materials from them can reduce our carbon footprint of the seats.

RHEON [8]

Rheon Labs - super absorbent lightweight polymer designed for high impact sports. Making use of this within aircraft seats will allow for a reduction in heavy structural material of the seats, and can increase effectiveness of reducing injuries in the event of an emergency.

Hydro

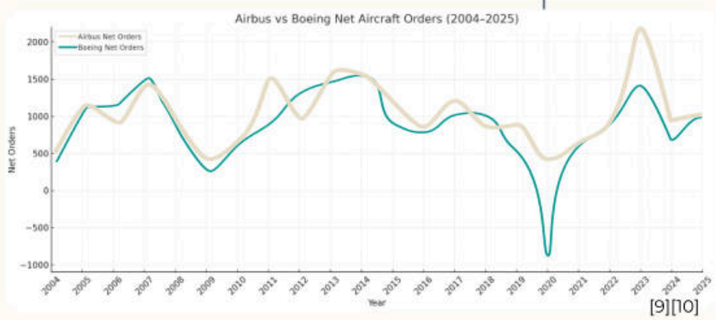
Hydro’s 4.0 aluminium is produced using 44% post-consumer recycled aluminium, making it a lower-carbon option suitable for facade systems- 6063 T6 aluminium, and a 95% reduction in energy required to produce the refined aluminium.

RECARO

SAFRAN

Seat Manufacturers: Recaro, Safran, Collins Aerospace.
There are two paths to go down- either start a new design company, or sell designs or the rights of the product to companies who already have a strong foothold in the market, and therefore guarantee revenue if these companies agree to work with us.

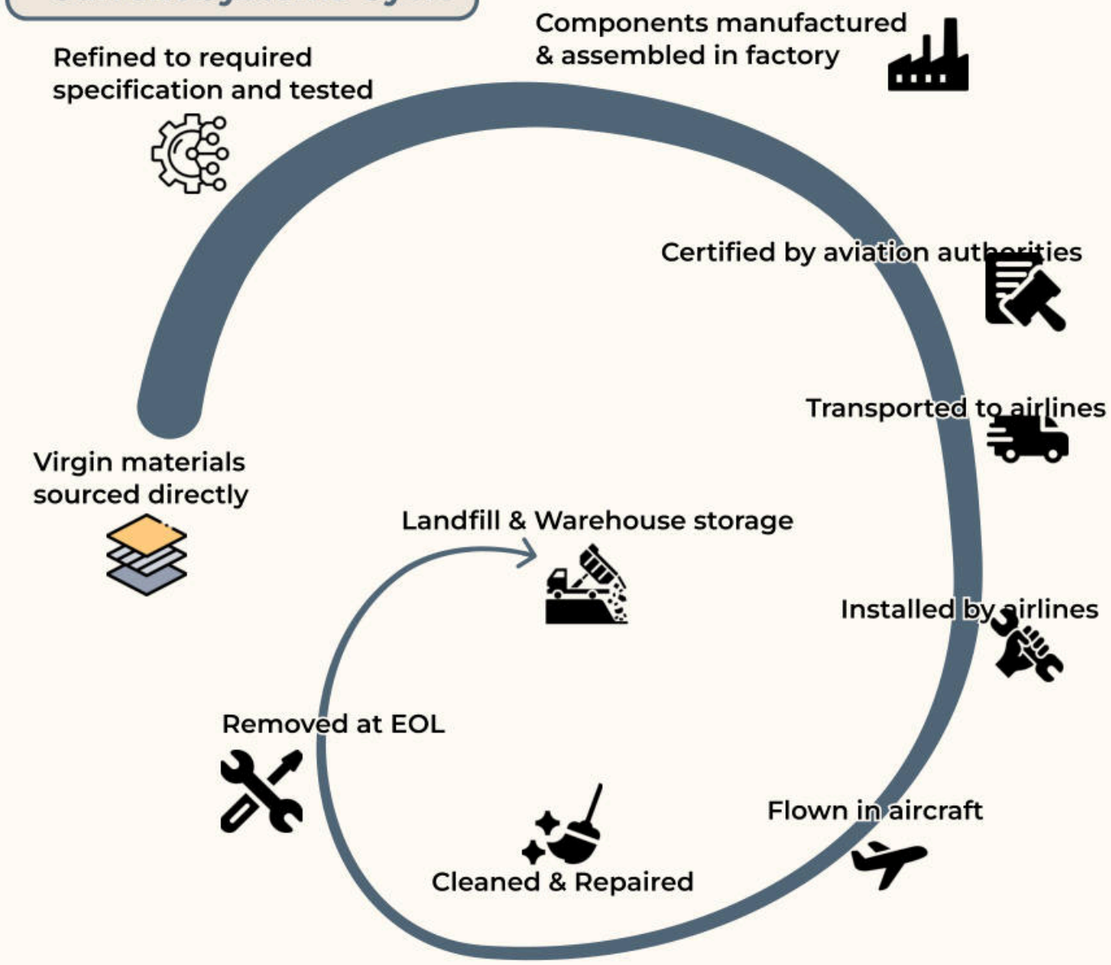
Collins Aerospace



[9][10]

Aside from global market crashes in 2008-09 and 2020, the aviation industry is clearly spiked- airlines will place mass orders, with the following years of them adjusting their options, followed by a drop in orders, as aircraft stay with airlines for typically 20 years

Current Systems Cycle

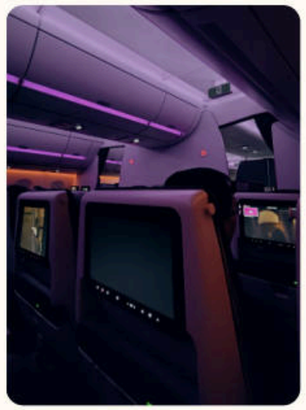


No circularity, no evidence of a circular economy nor a cyclic life cycle, as the seats end in landfill or warehouses for storage. Many gaps have been identified for improvements to the current industry approach:

- Use of newer materials within seats
- Parts tagged to store data on what material it is for post-operation
- Certified maintenance on seats
- Recover cosmetics for recycling
- Sell used seats on dedicated platform
- Full disassembly aided by assembly information from QR codes on structure
- Repurpose and recycle rest of the seat

Requirements & Benchmarks

Systems in place



Qatar Airways A350-900 economy. Cabin mood lighting- circadian rhythm & airline's colours.

Designed to reduce jetlag by helping passengers transition to the local time zone.

Currently implemented by aircraft type and options. A retrofit option could prove beneficial so that airlines do not need to rely on aircraft manufacturers nor wait for new ones to be delivered.

Emirates has launched two limited edition luxury bag collections, upon the refurbishment of their fleet, donating around $20,000 each time. A similar concept can be adopted to the cosmetic parts of the seats, before reselling the structural frame to other airlines, or participate in "buy-back" programs with seat manufacturers.



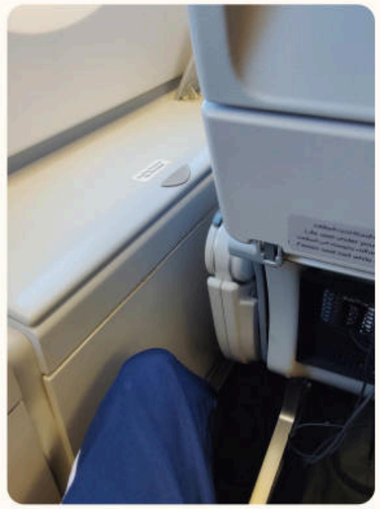
[11]

"Optiq provides two Bluetooth connections, built-in Wi-Fi for passengers to pair multiple devices including phones, tablets, headphones or even game controllers, while supporting up to 60 watts USB-C to fast-charge personal devices."

Emirates A350-900 uses Thales' AVANT Up Optiq 4k QLED HDR displays - the only aircraft displays using Samsung's QLED technology, which is offers some of the best visuals on the market. So there is a clear divergence happening regarding IFE, as British Airways is piloting BYOD (bring your own device) whilst Emirates is clearly enhancing options onboard. [12]

Requirements

Cosmetic, branded features of the seat must be easily removable for cleaning, repairs, and at the end of its service life with that airline.	Magnetically removable gel headrests- these undergo a lot of use as your head jostles about, wearing away the covering and cushioning.
Must weigh less than current seats on the market: <12kg	Ventilated seat pans with quick-release mechanism accessible beneath seat cover.
Redesigned seat must be comfier/ designed to be comfier than existing seats. Maintenance and cleaning should not degrade the product.	For example, covers secured with Velcro will have a greater tendency to peel off after repeated removal, due to the hooks in Velcro, or bleach removing dye. Digital twin of seat with sensors fitted to seats to monitor usage, measure stress points, and identify the best time for maintenance to avoid unnecessary inspections that increase downtime.
Every engineer responsible for seat disassembly of any seat from any company should be able to disassemble this seat easily.	QR codes for every parts and sub assembly are printed with links to disassembly for ease.
Entertainment system should be standard with seats featuring the latest technologies feasible for an aircraft.	Replace bulky, hard to reach plug sockets with 65W usb-c ports. display-capable usb c ports standardised on ife. Improve audio quality.
Last for 10 years with one airline, allowing for 2-3 rebrands before a new seat built on newer technologies is more favourable.	Tray table material must be reduced significantly, shifting towards a truss-braced table to support large items. Adjustable to accommodate varied sizes.
Shift focus to recycled materials rather than CFRPs to lengthen material life after use and reduce embodied energy.	Fix the source of the issue, and put systems in place to change the current strategy to align with a circular economy and improved life cycle of the product. turn a profit on the recycling of aircraft seats.
Reduce permanent fasteners to improve design flexibility & modularity for airlines, speed up inspections.	



Qatar Airways A380 upper deck economy, seat 27A

In this aircraft, the frame is too curved to add another seat, so the space was filled by deep storage bins, doubling as a side table. This meant that new seats in of themselves were not required to be installed here, but passengers still benefit from increased space.

When you sit in economy, especially on long flights, you hear or see people losing their belongings on the floor, often slipping between gaps in seats. Just as Steve Jobs dropped an iPod into a tank, watched bubbles escape and say "there's empty space, make it smaller", this means there are parts of the passenger's space that are yet to be utilised.

Seat Component	Primary Material	Sustainability Rationale	Key Properties	Example Suppliers
Seat Frame Structure	7075-T6 recycled aluminium alloy	High recycled content, low weight	High strength to weight, fatigue resistant	Constellium, Alcoa
Backrest Internal Structure	Recycled CFRP (20-40 percent)	Lightweight, reduced resin usage	High stiffness, customizable layups	Teijin, Toray
Seat Pan Base	Glass fiber composite with bio-resin	Lower embodied carbon, bio epoxy	High stiffness, thin shell shaping	Gurit, Hexcel
Energy Absorbing Zones	EPP foam	Fully recyclable, durable	Impact absorption, lightweight	JSP ARPRO
Cushion Foam	Bio-PU foam	Reduced petrochemical use	Comfort, fire treatable	Carpenter, Recticel
Seat Cover Fabric	Recycled polyester or wool blend	Recyclable, durable	Abrasion resistant, breathable	Camira, Lantal
Headrest Wings	Thin CFRP with rPET fabric	Slim, recyclable trim	High stiffness in thin sections	Toray, Lantal
Tray Table	Paper honeycomb with aluminium skins	Lightweight, recycled fiber	High stiffness to weight	Hexcel
Armrests	Bio-based PA11 or PA12	Renewable feedstock	High fatigue strength	Arkema Rilsan
Seat Track Fittings	Titanium Ti-6Al-4V	High recyclability	High strength, corrosion resistant	ATI, VSMPO
IFE Housing	Recycled PC-ABS	Recyclable, common in electronics	Impact resistant, mouldable	Sabic, Covestro
Underseat Shroud	Flax fiber + PP biocomposite	Renewable, lightweight	Good damping properties	Bcomp
Adhesives	Low-VOC polyurethane	Low toxins, easier recycling	Strong enough for trim	Henkel, 3M
Fasteners	Aluminium and titanium	Fully recyclable	Lightweight, corrosion resistant	PCC Fasteners



Opportunity: Implement online platform for maintenance

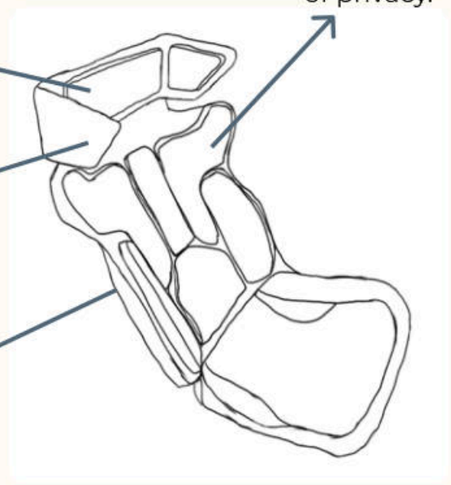
Next steps: Begin ideating designs for the physical product as well as the systems to implement.

Seat Ideation

Gel padding contours better than foam, so this is used for the headrest. Long-term deformation is not an issue as there are no heavy loads like that of the seat base acting upon it.

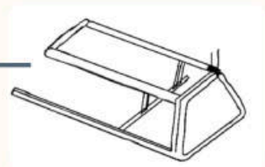
Arched sides for added visual sense of privacy.

Winged headrest is a step-up from current economy adjustable headrests, as it incorporates the padding with a physical barrier, adding to privacy. Few airlines have implemented this, and so far only in premium economy.



Galvorn seatback- Carbon Fibre alternative [16]

Floating style truss supports inspired by Greater Anglia Class 720 and 745 train seating- more space in the footwell as well reduced visual clutter.



Single piece of aluminium for strength and ease of assembly.

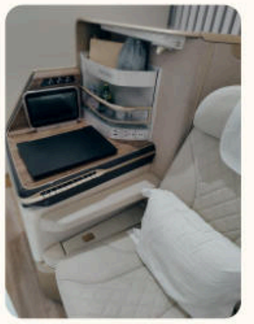
Digital windows for seats without a window- can be implemented in any aircraft allowing for a more consistent experience across passengers.

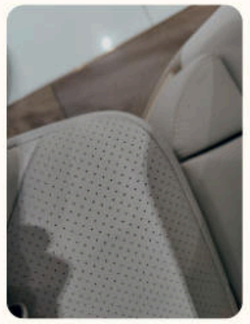


Double-decker style seating- for long haul flights, most luggage is checked-in, so two thirds of overhead lockers can be sacrificed for double stacking of seats.

Thigh support is crucial for good posture and comfort on long journeys, therefore implementing even a small thigh extension within the seat will make a huge difference to comfort on flights.

At the Emirates store, I was able to try out their premium products to draw some of their features to economy class- perforated cushions are there for breathability, but they could be connected to the aircraft's air conditioning/ heating to provide better climate control to the passenger.
There is also a usb-c port for power delivery as well as display connection capabilities.



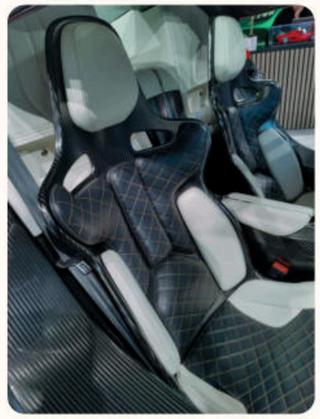




Etihad A321LR economy seats (photo taken in the Galleria Mall prior to official launch, may vary). Narrow-body plane but used on medium haul routes with long-haul-style seats. Seat base slides forward when reclining to respect other passengers' space, allowing you to recline guilt-free. This could be taken a step further by lowering the seat base to extend your legroom as you slide forward.

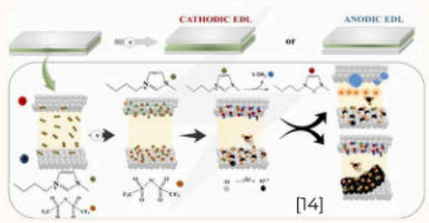
No seat gaps so the space is fully utilised and prevents belongings from falling in between.

Singular IFE box controls all 3 seats; can reduce to only in screen.

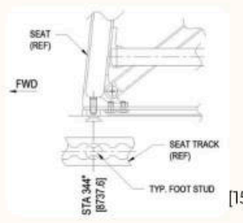


Koenigsegg Jesko Attack sports bucket seats are prioritised for racing over comfort, however are padded. Padding is only situated on areas crucial points of your body rest against, thus reducing weight. As weight reduction is central to the car, and Koenigsegg as a brand, this strategy can be confidently implemented within aircraft too.

For good posture, your hips, thighs, lower back and shoulders should be padded. With regular economy seats, the whole seat is padded. While marginally better, it will contribute to the weight a little, and make it potentially harder to clean.



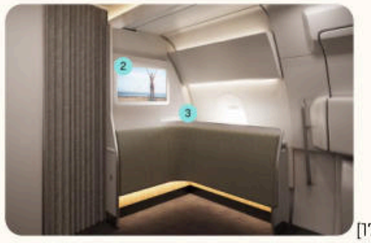
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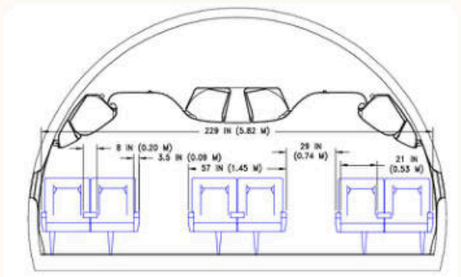
Electrically debondable adhesive- used in phone battery removals- if used to adhere fabrics to the metal frame, it can be quickly and easily unbonded during disassembly, allowing for better recycling and lowering labour costs.

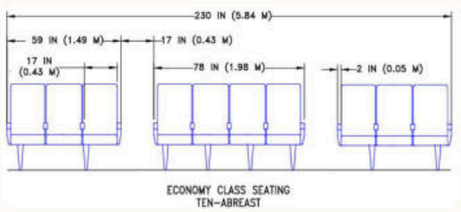
Systems Ideation

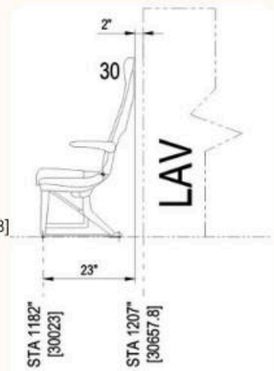
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Qantas A350-1000ULR Project Sunrise uses a “wellbeing zone” on the aircraft for passengers to stretch during the upcoming 21 hour flight, open to all passengers on the flight. Losing this much space is reflected in the price of tickets, however it is an interesting system to adopt. Saudia similarly gives up a large space but for people to pray on their 777s. These kinds of implementations add arguably far more than any advancement in an economy seat could offer, so these breakout zones could be coupled with cheaper, less comfortable seating to outweigh the comfort, whilst reducing weight (albeit passengers as well).





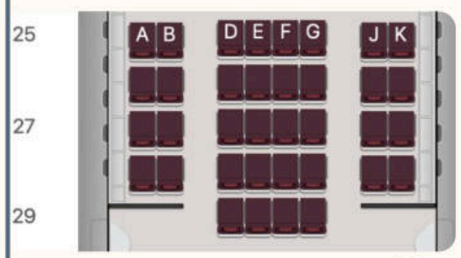
Restrictions



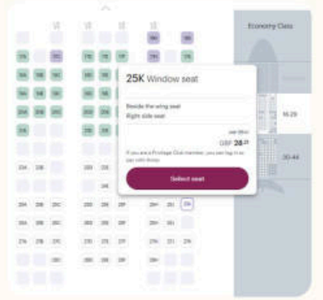
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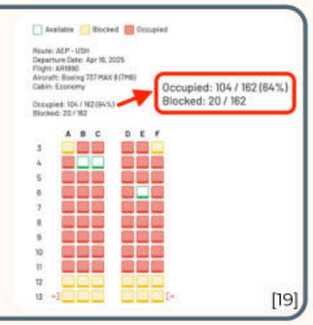
Based on typical widebody aircraft interior dimensions, the seating for my economy class will be designed for 9-abreast, with modular connectors on either side of end seats, to allow for 1-2, or even 3 middle seats to be added, allowing airlines to install the same seat product in any of their widebodies from the 787-8 to the A350-1000.
Seats that slide forward also combat the restricted recline of seat in front of lavatories.

Digital Tools



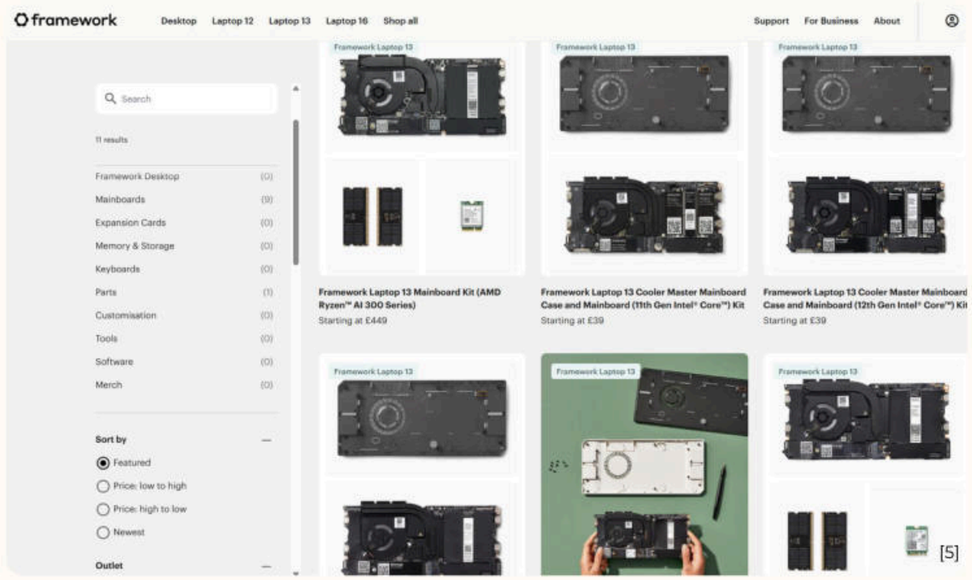
[4]





[19]

As pictured on the left, some seats align well with windows like 28A/K, but 26A/K mostly misses out on the window, and has an awkward storage bin placement. Paying the same amount of money for vastly different products is not right, and should be made more apparent when choosing seats on airline websites. I have included an example of what seat selection looks like for an upcoming flights- which actually costs £20-£100 just pre-select a seat for.



[5]

Certified marketplace for the sale of used airline seats- mainstream **airlines can recuperate some money** whilst budget airlines save.

Retrofit categories introduced to **add new features to ageing aircraft**, such as circadian lighting, controllable AC and dimmable windows.

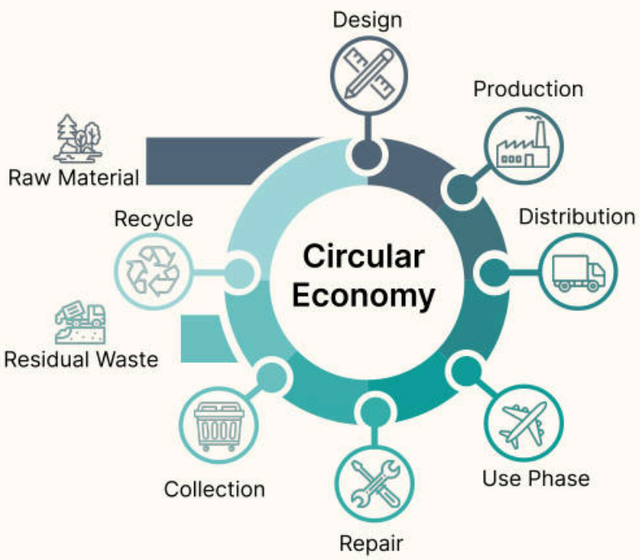
Design changes aligned with **brand storytelling** can enhance perceived value for consumers, a **key** driver of **financial success.**

Design decisions are driven by regulators, manufacturers, and airlines prioritising **compliance, efficiency, and profit.**

Operational staff are underrepresented in early design stages.
Consider systems that **help groundstaff work with the plane better.**

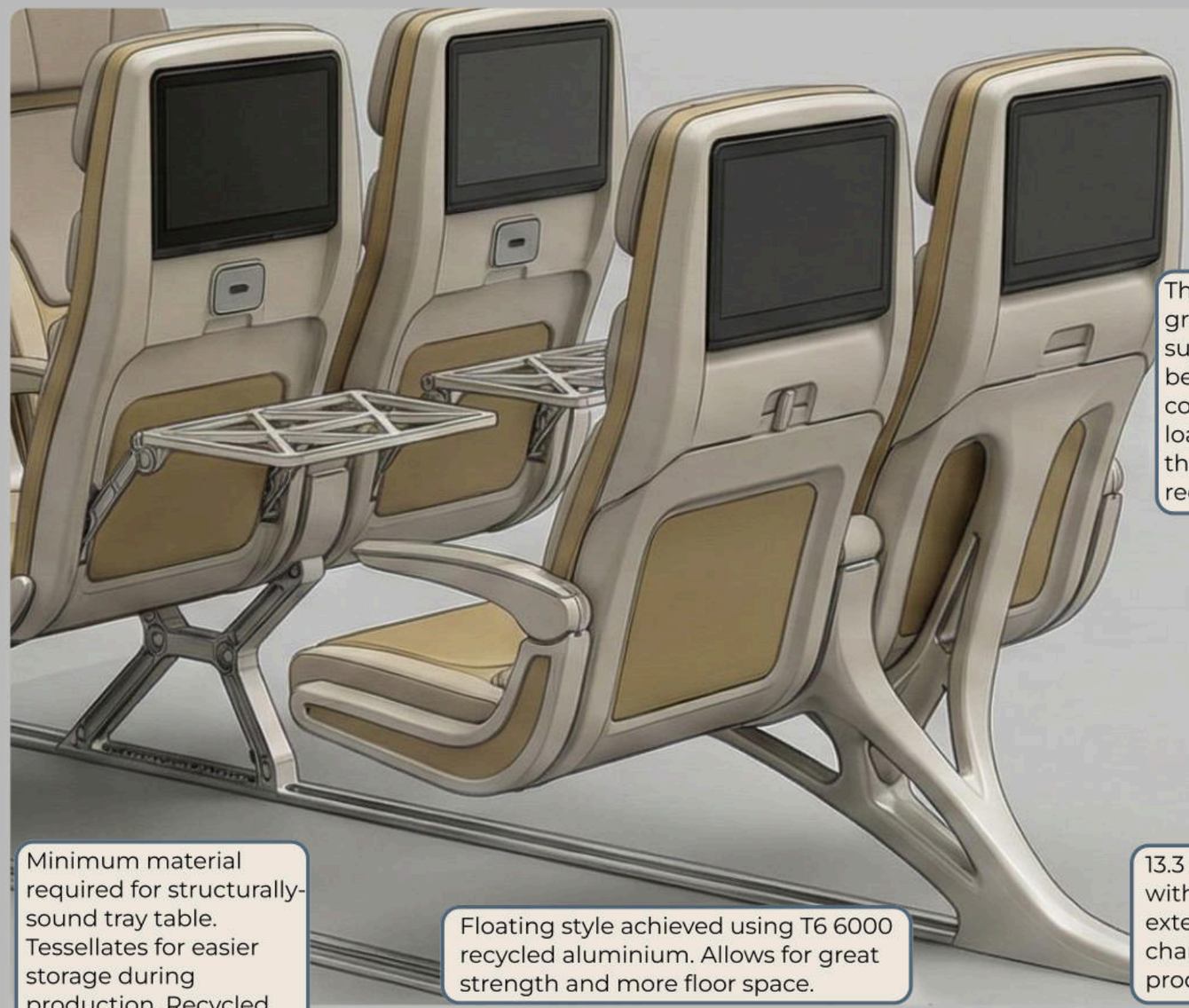
Seat manufacturers don’t control high-level design decisions, but they can launch **buy-back programs** to refurbish/sell on used seats for others.

Systems Thinking




Winged privacy headrest
Sectioned padding for targeted essential comfort
Perforated fabrics & leather for ventilation
Reduced volume frame with 1 fastener

Exploded Diagram



Thigh extension for greater leg support, for a better posture and comfort. Spring loaded and works the same way as reclining the seat.

Minimum material required for structurally-sound tray table. Tessellates for easier storage during production. Recycled GFRP.

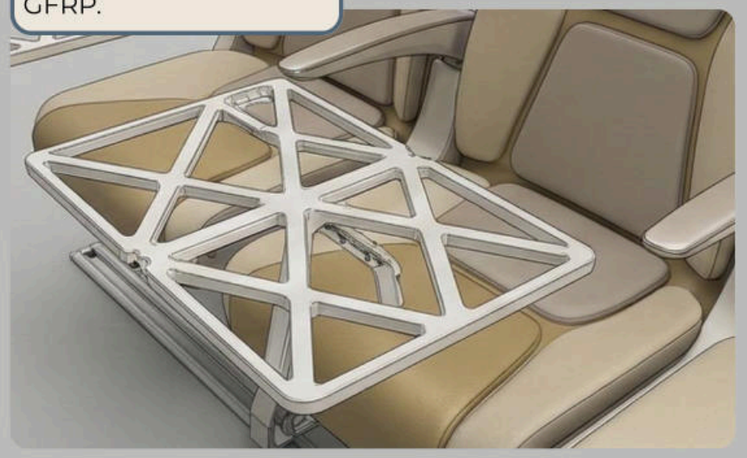
Floating style achieved using T6 6000 recycled aluminium. Allows for great strength and more floor space.

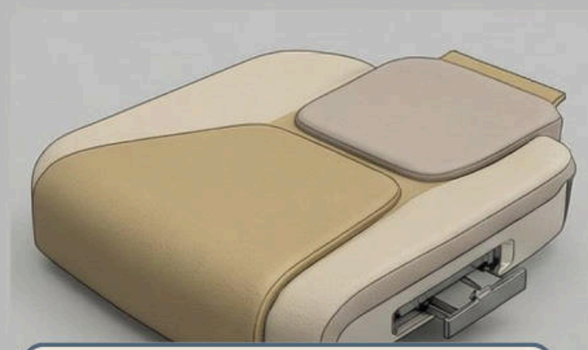


13.3 inch 4K oled touch screens with IFE built-in, eliminating an external box. Usb-c port for fast charging and screen sharing for productivity or entertainment.

3 bolts used per seat, with channels cut into mostly uni-bodied pieces of aluminium to reuduce weight and ease disassembly process.

Padding separated based on important contact points with body for optimised comfort.





Mechanically fixed seat base- no fasteners required as it latches in and locks with springs to secure itself.



Systems Re-Mapping

Existing system

Proposed new system

Transport

Energy-intense process

Refinement of resources

Virgin resource extraction

Limiting this as much as possible through feedback from recycled components & repurposing.

Production of specific materials and components for the airline seat

RECARO

Certification from FAA/CAA/EASA

Civil Aviation Authority

Inspections

Digital twin monitors usage and condition requiring less physical maintenance.

Cleaning

Ground staff will remove the seat cushions and deep clean them.

Failed inspection- broken part or no longer air-worthy

Refer to the QR codes on the parts for disassembly.

BRITISH AIRWAYS

Usage in flight

Installation in aircraft

Refer to the QR codes on the parts for installation.

Re-introduction of repaired component

Hydro

Recycle parts of product

Disassembled

Refer to the QR codes on the parts for disassembly.

Heavy duty transportation to landfill- emissions beyond our control.

Landfill or incineration

Further disassembly

Gen Phoenix

Repurpose

Gen Phoenix and dedicated tailor shops can repurpose fabrics and leather for consumer goods or recycled materials.

Sustainability

90% materials recycled

As a result of collaboration with multiple stakeholders & new design.

Customer

No utensil wastage as they are eaten with the meal- reduces single-use plastics.

Cost-savings

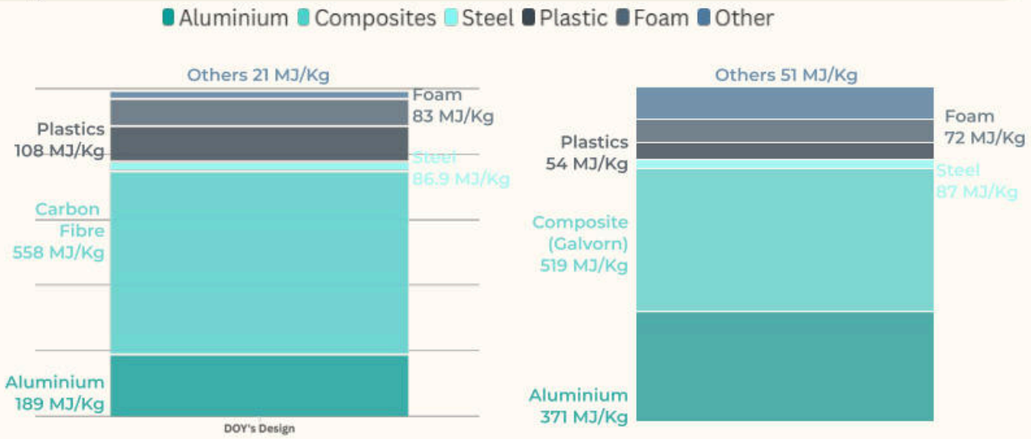
Airline branding made simple through modular refurbishments & resell market.

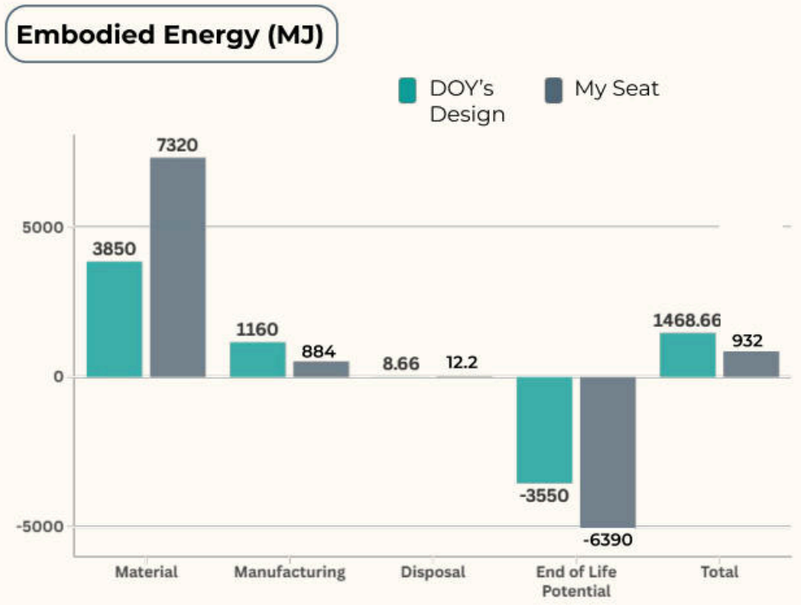
Accurate checks of structural integrity reduce downtime and lengthen product service life.

Environmental Evaluation & Value Proposition

Materials Embodied Energy per Kg

My seat design compared to DOY Design using Ansys Eco Audit tool, demonstrates stark differences in approach; electing to reduce carbon fibre usage.





My seat weighs 20% more than the Doy Design seat, and about 4% more than leading seats on the market such as Recaro CL3810. However, it has a lower embodied energy, and the reduction in hard-to-recycle materials allows my seat to be recycled more easily and to a greater extent, lowering its carbon footprint at EOL. Although this will increase operating costs slightly, it will vastly reduce maintenance and repair costs, as it is more resistant to vibrational forces, even when not considering the sensor implementations to monitor the seat.

Value Proposition & Next Steps

In-flight experience

- As technology advances, better and bigger screens can be implemented.
- Airlines are currently rolling out Starlink WiFi- free fast internet (250mbps) from gate-to-gate may negate the need for IFE entirely.
- As cabin pressurisation improves, food will start tasting better.

Partnerships

- Reach out to these stakeholders and submit a formal proposal with the entire plan and process laid out, speak to regulators, and look for investors to start the company.
- Work closely with GenPhoenix, RHEON Labs and Galvorn for material selection process.
- Secure a factory for production.

Seat Comfort

- Improved passenger comfort whilst sitting.
- Bigger IFE screen and screenshare capabilities boost productivity and entertainment options.
- Area will feel less cluttered to passengers, leading to fewer complaints and a better airline ranking.
- Ample legroom beneficial to all passengers.

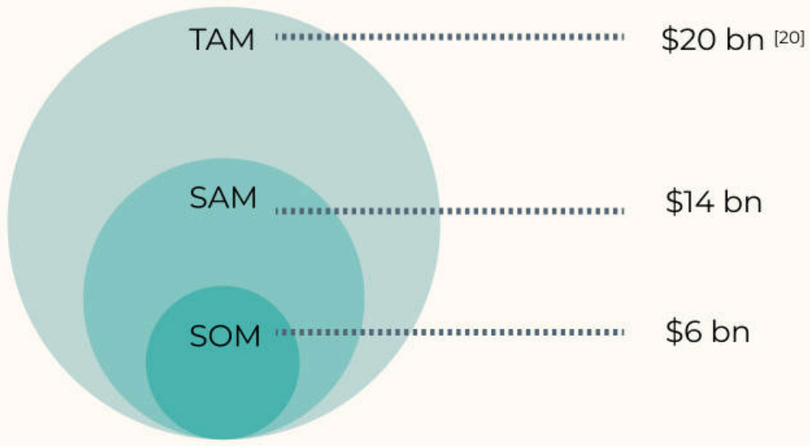
Groundstaff operations

- Cleaning crew can more easily remove seat covers for deeper cleaning, speeding up their work and decreasing down-time for aircraft.
- Maintenance crew are able to disassemble seats at a profitable pace, freeing up warehouses full of old seats.

Politics

- Government subsidies will take time to implement, but with time these can be incorporated into seat design, as well as perhaps subsidising locally produced parts for seats geared towards sustainability- helping to keep UK primary manufacture economies running whilst passing on the benefits to the airlines and manufacturers.
- Socially this redesign should be welcomed if its impact can be demonstrated and made clear, not just about lowering the carbon footprint through system changes, but also by encouraging the recycling of it.

Market Size



Appendix

AI_generated content used for Hero image and Exploded diagram based on my drawings, research and tailoring.
AI also used to create the images of the 4 personas in the “Key Users”, indicated y the ai logos in the bottom corners.

1. https://aviationbenefits.org/faqs/electric-aircraft/#:~:text=Other%20similar%20sized%20aircraft%20are,and%20how%20do%20they%20work?
2. https://www.jetzero.aero/aircraft
3. https://www.skyart.com/airbus-aircraft-crew-seat?srsltid=AfmBOopfV5dYE18yKypeH3Lru-TlHkrMsWjS_2BzzROTDTEw-9pHax19
4. https://www.aerolopa.com/qr-388
5. https://frame.work/gb/en/try-it-out
6. https://incredibleeats.com/?srsltid=AfmBOooHqRnOikr1QPMK1TkTmUMfTQsjZWfvO7EwuqsCrj4YKXehIuYX
7. https://genphoenix.com/
8. https://rheonlabs.com/
9. https://investors.boeing.com/investors/overview/default.aspx
10. https://www.airbus.com/en/products-services/commercial-aircraft/orders-and-deliveries[13]
11. https://www.emirates.com/media-centre/emirates-drops-second-limited-edition-collection-of-luggage-made-from-upcycled-aircraft-interiors---aircrafted-by-emirates/
12. https://www.emirates.com/media-centre/emirates-invests-over-us--350-million-in-next-generation-inflight-entertainment-systems-for-new-a350-fleet/
13. https://www.qantas.com/gb/en/about-us/our-company/fleet/new-fleet/project-sunrise.html
14. https://www.sciencedirect.com/science/article/abs/pii/S2352492823009844
15. https://pmc.ncbi.nlm.nih.gov/articles/PMC9952460/
16. https://dexmat.com/blog/carbon-fiber-alternative/
17. https://www.qantas.com/cn/en/about-us/our-company/fleet/new-fleet/project-sunrise/wellbeing-zone.html
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19. https://blog.awardfares.com/seatmaps/
20. https://www.grandviewresearch.com/industry-analysis/aircraft-seating-market-report

All icons: https://www.flaticon.com/